Class 18: Pertussis and the CMI-PB project

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Background

Pertussis (aka Whooping Cough) is a common lung infection caused by the bateria B. Pertussis.

This can infect people of any age but it is most severe and deadly against infants.

The CDC track the number of reported cases in the US

We can "scrape" this data with the **datapasta** package!

```
cdc <- data.frame(
  year = c(
    1922L,
    1923L,
    1924L,
    1925L,
    1926L,
    1927L,
    1928L,
    1929L,
    1930L,
    1931L,
    1932L,
    1933L,
    1934L,
```

```
    1935L,
    1936L,
    1937L,
    1938L,
    1939L,
    1940L,
    1941L,
    1942L,
    1943L,
    1944L,
    1945L,
    1946L,
    1947L,
    1948L,
    1949L,
    1950L,
    1951L,
    1952L,
    1953L,
    1954L,
    1955L,
    1956L,
    1957L,
    1958L,
    1959L,
    1960L,
    1961L,
    1962L,
    1963L,
    1964L,
    1965L,
    1966L,
    1967L,
    1968L,
    1969L,
    1970L,
    1971L,
    1972L,
    1973L,
    1974L,
    1975L,
    1976L,
```

```
    1977L,
    1978L,
    1979L,
    1980L,
    1981L,
    1982L,
    1983L,
    1984L,
    1985L,
    1986L,
    1987L,
    1988L,
    1989L,
    1990L,
    1991L,
    1992L,
    1993L,
    1994L,
    1995L,
    1996L,
    1997L,
    1998L,
    1999L,
    2000L,
    2001L,
    2002L,
    2003L,
    2004L,
    2005L,
    2006L,
    2007L,
    2008L,
    2009L,
    2010L,
    2011L,
    2012L,
    2013L,
    2014L,
    2015L,
    2016L,
    2017L,
    2018L,
```

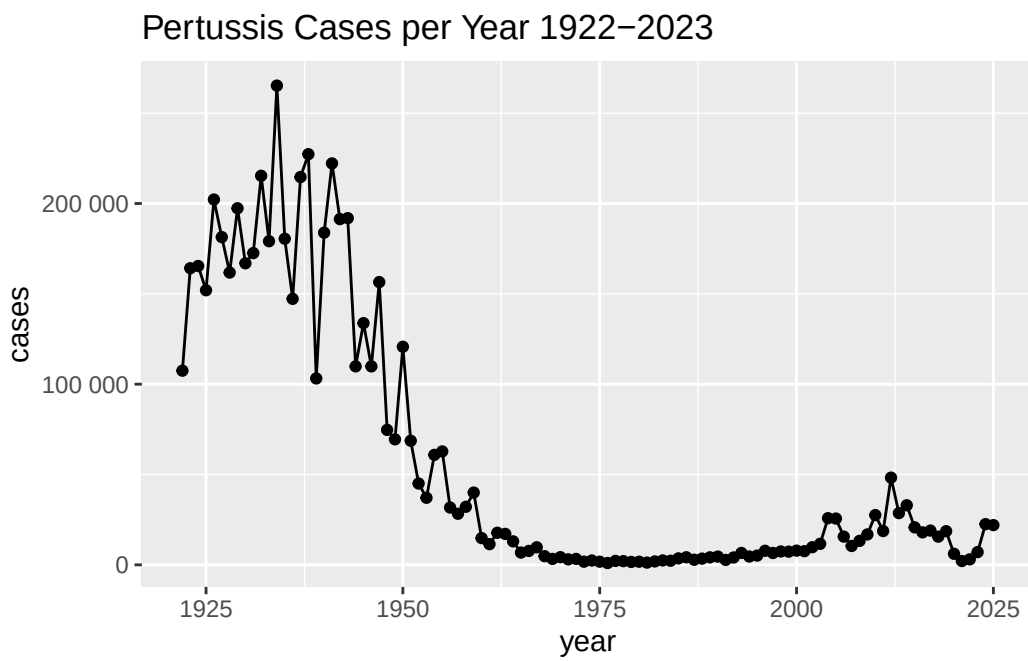
```
    2019L,
    2020L,
    2021L,
    2022L,
    2023L,
    2024L,
    2025L
  ),
  cases = c(
    107473,
    164191,
    165418,
    152003,
    202210,
    181411,
    161799,
    197371,
    166914,
    172559,
    215343,
    179135,
    265269,
    180518,
    147237,
    214652,
    227319,
    103188,
    183866,
    222202,
    191383,
    191890,
    109873,
    133792,
    109860,
    156517,
    74715,
    69479,
    120718,
    68687,
    45030,
    37129,
    60886,
```

```
    62786,
    31732,
    28295,
    32148,
    40005,
    14809,
    11468,
    17749,
    17135,
    13005,
    6799,
    7717,
    9718,
    4810,
    3285,
    4249,
    3036,
    3287,
    1759,
    2402,
    1738,
    1010,
    2177,
    2063,
    1623,
    1730,
    1248,
    1895,
    2463,
    2276,
    3589,
    4195,
    2823,
    3450,
    4157,
    4570,
    2719,
    4083,
    6586,
    4617,
    5137,
    7796,
```

```
    6564,
    7405,
    7298,
    7867,
    7580,
    9771,
    11647,
    25827,
    25616,
    15632,
    10454,
    13278,
    16858,
    27550,
    18719,
    48277,
    28639,
    32971,
    20762,
    17972,
    18975,
    15609,
    18617,
    6124,
    2116,
    3044,
    7063,
    22530,
    21996
  )
)
```

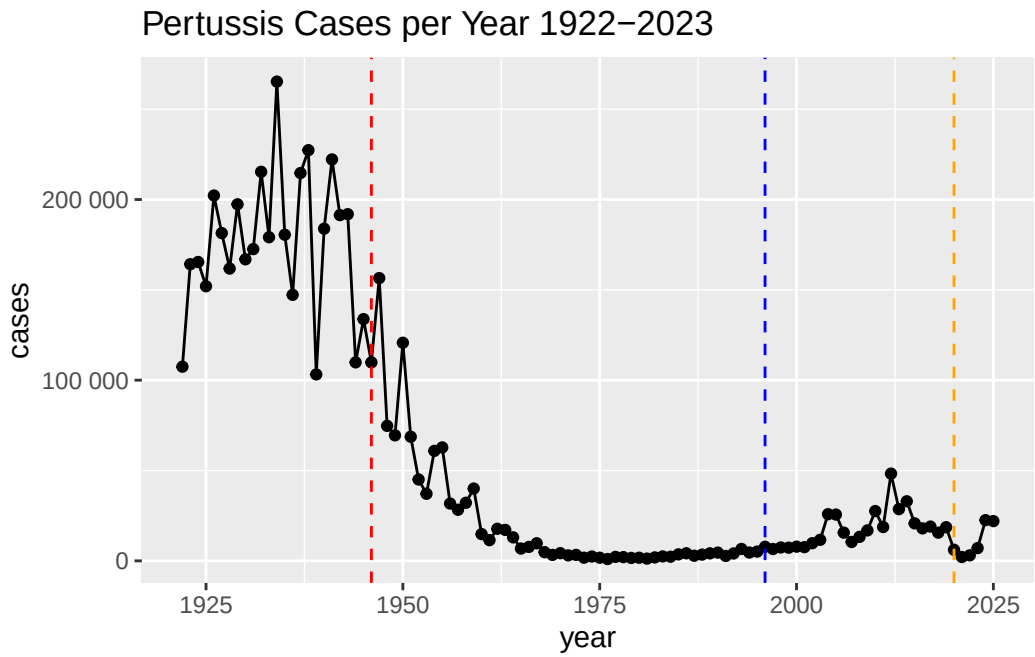
Q1. Make a plot of `year` vs `cases`

```
library(ggplot2)
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  labs(title = "Pertussis Cases per Year 1922-2023") +
  scale_y_continuous(labels = scales::label_number())
```



Q. Add some major milestones including the first wP vaccine rollout (1946), the switch to the newer aP vaccine (1996), and the Covid years (2020).

```
library(ggplot2)
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  labs(title = "Pertussis Cases per Year 1922-2023") +
  scale_y_continuous(labels = scales::label_number()) +
  geom_vline(xintercept = 1946, color="red", linetype = "dashed") +
  geom_vline(xintercept = 1996,color = "blue", linetype = "dashed") +
  geom_vline(xintercept = 2020,color = "orange", linetype = "dashed")
```



There we high case numbers in the pre-1940s as a result of no available vaccines. Following 1946, the release of the wP vaccine was widely distributed leading to a dramatic decrease in cases. In 1996, the new aP vaccine was release which didn't really come with a decrease in cases. We see the rise of cases once again in 2004 which may have been likely to the waning immunity caused from the vaccines. Additionally the anti-vax movement and political landscapes may have decreased the amount of people getting the vaccine. Another factor could be that our testing was getting better hence the increased amount of cases. In 2020, we see a dramatic decrease in cases due to the isolation and social distance guidelines put in place which prevented respiratory based illnesses.

Why is this vaccine-preventable disease on the upswing? To answer this question we need to investigate the mechanisms underlying waning protection against pertussis. This requires evaluation of pertussis-specific immune responses over time in wP and aP vaccinated individuals.

CMI-PB Project

Computational Models of Immunity - Pertussis Boost aims to provide the scientific community with this very information.

They make there data available via JSON format API. We an read this in R with the `read_json()` function from the **jsonlite** package

```r
library(jsonlite)

subject <- read_json("http://cmi-pb.org/api/v5_1/subject", TRUE)

head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

Q. Hw many wP and aP individuals area in this `subject` table

```r
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q. What is the biological sex breakdown?

```r
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

Q. In terms of race and gender is this dataset representitive of the US population?

```r
table(subject$race, subject$biological_sex)
```

```
                                          Female Male
  American Indian/Alaska Native                0    1
  Asian                                       32   12
  Black or African American                    2    3
  More Than One Race                          15    4
  Native Hawaiian or Other Pacific Islander    1    1
  Unknown or Not Reported                     14    7
  White                                       48   32
```

This dataset is not representitve of the US population as it was taken during Covid.

Let's read some more database tables!

```r
specimen <- read_json("https://www.cmi-pb.org/api/v5_1/specimen", TRUE)

ab_titer <- read_json("https://www.cmi-pb.org/api/v5_1/plasma_ab_titer", TRUE)
```

```r
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

```r
head(ab_titer)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

To analyze this data we need to first "join" (merge/link) the different tables so we have all the data in one place not spread across different tables.

We can use the `*_join()` family of functions from the **dplyer** to do this.

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
abdata <- inner_join(ab_titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(abdata, 5)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 UG/ML                 2.096133          1          wP         Female
2 IU/ML                29.170000          1          wP         Female
3 IU/ML                 0.530000          1          wP         Female
4 IU/ML                 6.205949          1          wP         Female
5 IU/ML                 4.679535          1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
```

Q. What Antibody isotypes are measured for these patients?

```r
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4 
 6698  7265 11993 12000 12000 12000 
```

Q. What antigens are reported?

```r
table(abdata$antigen)
```

```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA 
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318 
    PD1     PRN      PT     PTM   Total      TT 
   1970    6712    6712    1970     788    6318 
```

> Q12. What are the different $dataset values in abdata and what do you notice about the number of rows for the most "recent" dataset?

```r
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset 
       31520         8085         7301        15050 
```

Most recent data set has a lot more data compared to the data sets from 2021-2022. However, it is significantly less than the 2020 data set.

Lets focus on IgG antigen and make a plot of MFI_normaized for all antigens.

```r
igg <- abdata |>
  filter(isotype =="IgG")

head(igg)
```

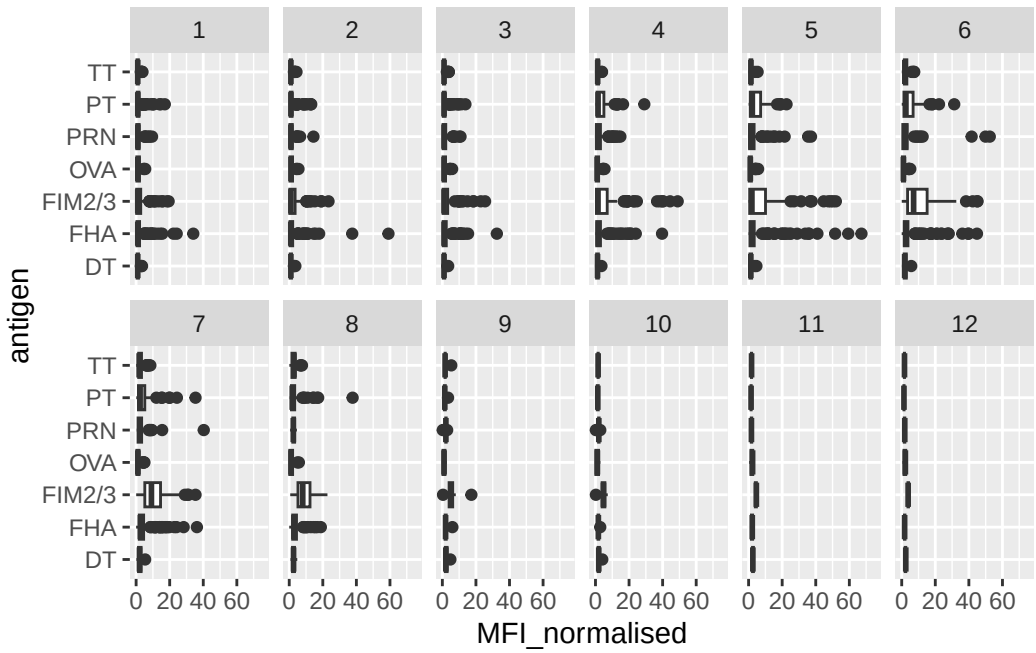
```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised 
1           1     IgG                TRUE      PT   68.56614       3.736992 
2           1     IgG                TRUE     PRN  332.12718       2.602350 
3           1     IgG                TRUE     FHA 1887.12263      34.050956 
4          19     IgG                TRUE      PT   20.11607       1.096366 
5          19     IgG                TRUE     PRN  976.67419       7.652635 
6          19     IgG                TRUE     FHA   60.76626       1.096457 
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 IU/ML                 0.530000          1          wP         Female
2 IU/ML                 6.205949          1          wP         Female
3 IU/ML                 4.679535          1          wP         Female
4 IU/ML                 0.530000          3          wP         Female
5 IU/ML                 6.205949          3          wP         Female
6 IU/ML                 4.679535          3          wP         Female
```

```
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4                Unknown White    1983-01-01    2016-10-10 2020_dataset
5                Unknown White    1983-01-01    2016-10-10 2020_dataset
6                Unknown White    1983-01-01    2016-10-10 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

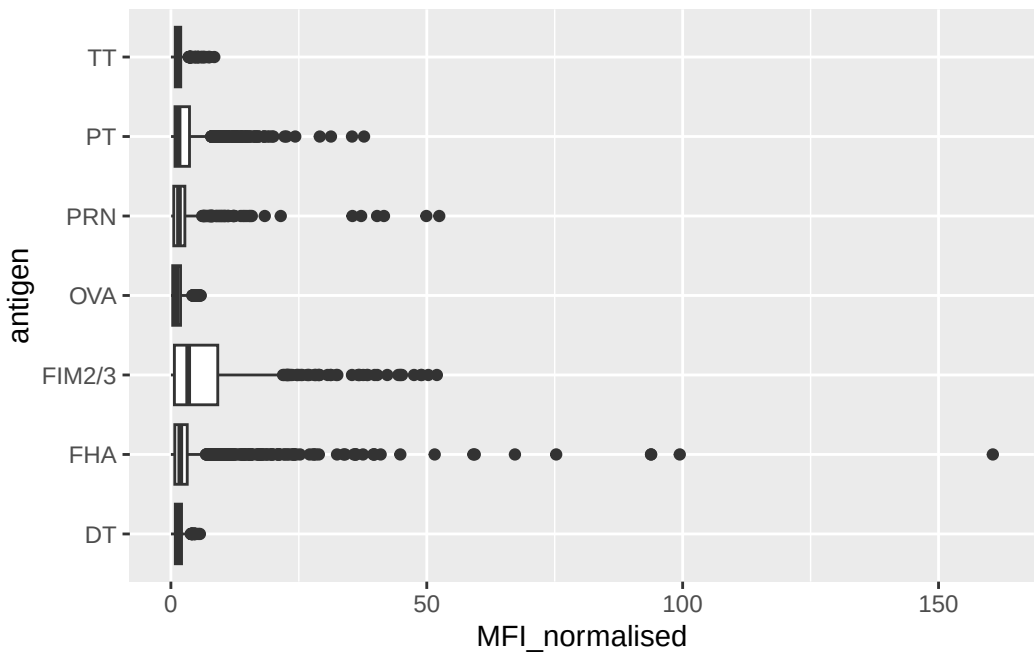
Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```r
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
    xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```

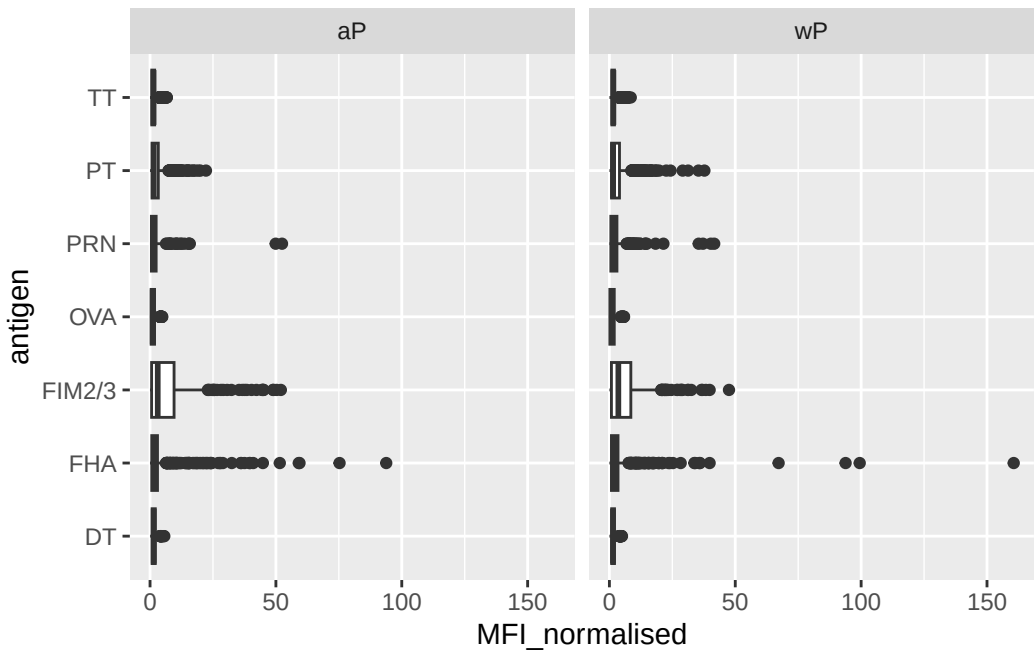



```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```

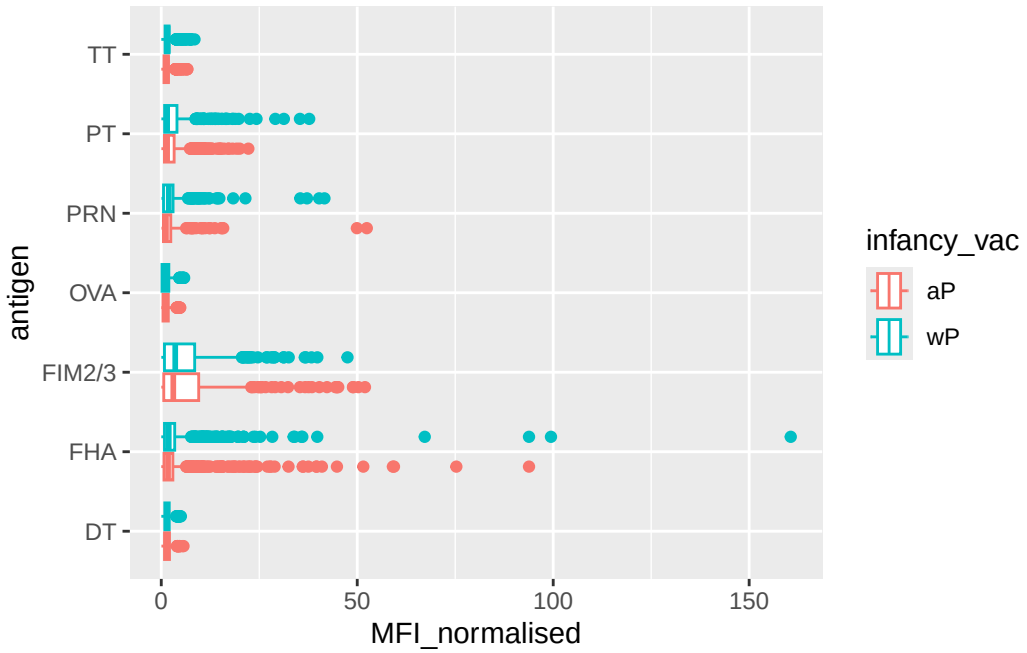


Q. Is there a difference of aP vs wP individuals with these values?

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```

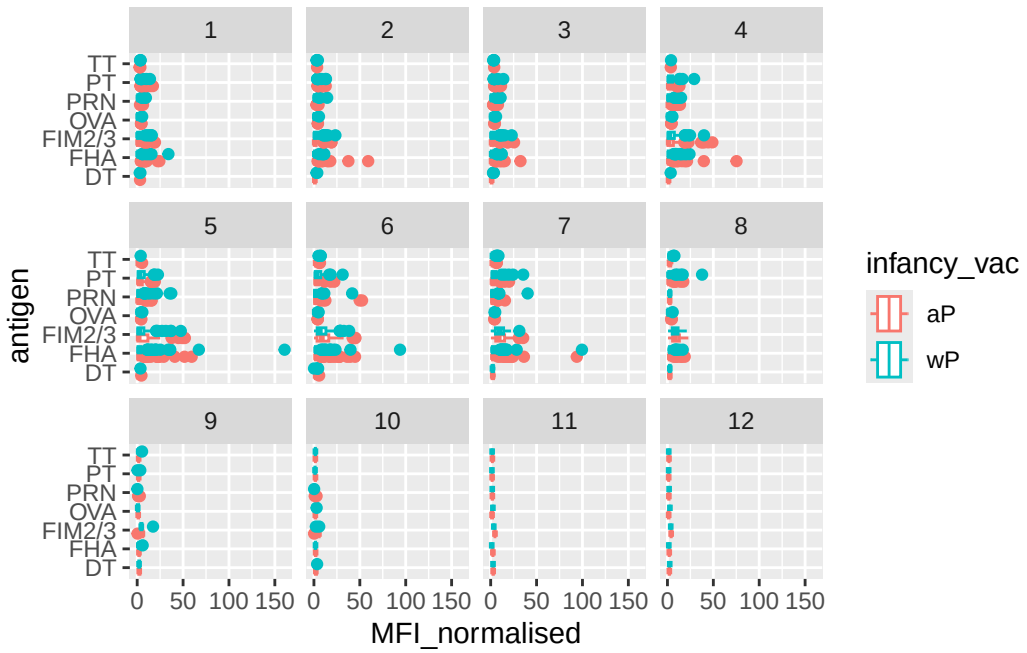



```
ggplot(igg) +
  aes(MFI_normalised, antigen, color = infancy_vac) +
  geom_boxplot()
```



Q. Is there a temprol response ie. do values increase or decrease over time?

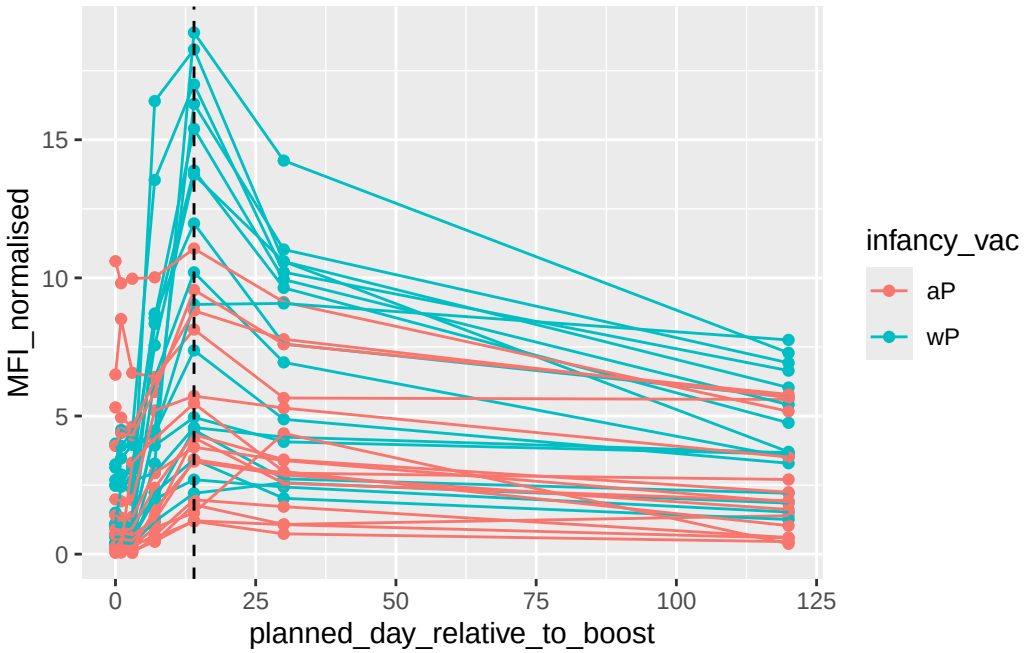
```
ggplot(igg) +
  aes(MFI_normalised, antigen, color = infancy_vac) +
  geom_boxplot() +
  facet_wrap(~visit)
```



Focus on "PT" Pertusisis toxin antigen

```
pt.igg21 <- igg |> filter(antigen =="PT",
              dataset == "2021_dataset")
```

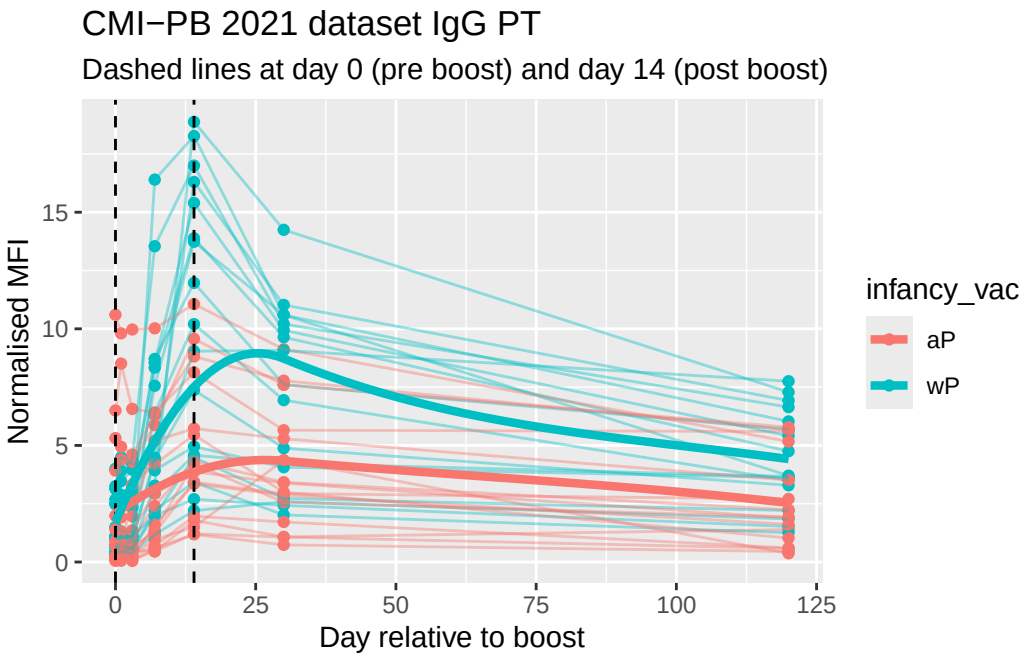
```
ggplot(pt.igg21) +
  aes(planned_day_relative_to_boost, MFI_normalised, col = infancy_vac, group = subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 14, col="black",lty="dashed")
```




```
ggplot(pt.igg21) +
  aes(planned_day_relative_to_boost, MFI_normalised, col = infancy_vac, group = subject_id) +
  geom_point() +
  geom_line(alpha = 0.4) +
  geom_vline(xintercept = 14, col="black",lty="dashed") +
  geom_vline(xintercept = 0, col="black", lty="dashed") +
  geom_smooth(aes(group = infancy_vac), method = "loess", span = 1.06, se = FALSE, linewidth

  labs(title = "CMI-PB 2021 dataset IgG PT",
       subtitle = "Dashed lines at day 0 (pre boost) and day 14 (post boost)",
       x = "Day relative to boost",
       y = "Normalised MFI")
```

```
`geom_smooth()` using formula = 'y ~ x'
```


CMI−PB 2021 dataset IgG PT
Dashed lines at day 0 (pre boost) and day 14 (post boost)
Normalised MFI
15
10
5
0
0
25
50
75
100
125
Day relative to boost
infancy_vac
aP
wP